

Cheat Sheet Materi 2

TI5001 / Optimisasi & Intelegensia Komputasional · Dasar-Dasar Kalkulus dan Aljabar

1 · Turunan: definisi & aturan

$df/dx = \lim_{\Delta x \rightarrow 0} [f(x+\Delta x) - f(x)] / \Delta x$
= kemiringan (slope) f di titik x. Notasi: $df/dx = f'(x)$

Aturan	Turunan
konstanta c	0
pangkat x^n	$n \cdot x^{n-1}$
$g \pm h$	$g' \pm h'$
perkalian $g \cdot h$	$g'h + g \cdot h'$
pembagian g/h ($h \neq 0$)	$(g'h - g \cdot h') / h^2$
rantai $y=f(u)$, $u=g(x)$	$dy/dx = (dy/du)(du/dx)$

Tabel slide: $(e^x)' = e^x$, $(\ln x)' = 1/x$, $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$

- $x^2(x+2) \rightarrow 2x(x+2) + x^2 \cdot 1 = 3x^2 + 4x$
- $x^2/(x+2) \rightarrow (x^2+4x)/(x+2)^2$; di $x=1$: **5/9**
- $y=u^2$, $u=x+2 \rightarrow 2u = 2x+4$
- $f=x^3$: $f'=3x^2$, $f'(1)=3$ (slope di $x=1$)

Perkalian: jangan mengalikan turunan dengan turunan.

2 · Orde tinggi, parsial, gradien

- Orde 2: $d^2f/dx^2 = d/dx(df/dx)$. $x^3 \rightarrow 3x^2 \rightarrow 6x \rightarrow 6$
- Parsial** $\partial f/\partial x_i$: turunan terhadap x_i , variabel lain dianggap konstan
- Campuran: $\partial^2 f/\partial x_i \partial x_j = \partial^2 f/\partial x_j \partial x_i$ (urutan bebas)

$f = x_1^3 + 2x_2^2 + x_1^2 x_2$: $\partial f/\partial x_1 = 3x_1^2 + 2x_1 x_2$; $\partial f/\partial x_2 = 4x_2 + x_1^2$; campuran = **$2x_1$**

Gradien ∇f = vektor semua turunan parsial pertama. $f = x_1 x_2^2 + x_2 x_3$ di $(1,2,3)$: $(x_2^2, 2x_1 x_2 + x_3, x_2) = \mathbf{(4, 7, 2)}$

3 · Matriks & operasi

- Ukuran $m \times n$ (m baris, n kolom). **Transpos** A^T : baris $\leftrightarrow$ kolom, ukuran $n \times m$
- $A \pm B$: ukuran harus sama; kA : tiap elemen $\times k$
- AB**: $(m \times k)(k \times n) \rightarrow m \times n$; kolom A = baris B; elemen $(i,j) = \sum_p a_{ip} b_{pj}$
- Identitas I ($n \times n$): diagonal 1, lainnya 0

$A \ 3 \times 2$, $B \ 2 \times 4$, $C \ 3 \times 4$: $AB \ 3 \times 4 \rightarrow \mathbf{AB+C}$ ada; BA tak terdefinisi ($4 \neq 3$); $A^T B$ tak terdefinisi ($3 \neq 2$)

$$\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ -1 & -4 \end{bmatrix}$$
$$\begin{bmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 \\ -2 & 0 \\ -1 & 2 \end{bmatrix}$$

4 · Determinan, minor, kofaktor

Ukuran	det A
1x1	a_{11}
2x2	$a_{11}a_{22} - a_{12}a_{21}$
3x3	$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$

- Minor** M_{ij} : hapus baris i & kolom j

- Kofaktor** $c_{ij} = (-1)^{i+j} \det M_{ij}$

$\det A = \sum_j a_{ij} c_{ij}$ (baris i tertentu) = $\sum_i a_{ij} c_{ij}$ (kolom j tertentu)

$$A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 0 & 2 \\ 1 & -1 & -2 \end{bmatrix} \rightarrow c_{11}=2, c_{12}=0, c_{13}=1$$

$$\det A = 1 \cdot 2 + (-2) \cdot 0 + (-1) \cdot 1 = 1$$

$$2 \times 2 \begin{bmatrix} 1 & -2 \\ -1 & 0 \end{bmatrix}: 1 \cdot 0 - (-2)(-1) = -2$$

5 · Principal & leading principal minor

- Principal minor orde k**: submatriks dari menghapus $(n-k)$ baris dan kolom dengan indeks sama $\rightarrow$ total $2^n - 1$
- Leading principal minor orde k**: hapus $(n-k)$ baris & kolom terakhir $\rightarrow$ tepat **n** buah

			
leading orde 1	leading orde 2 principal, bukan leading		orde 3 = det A

$n=3$: 7 principal minor, 3 leading. $n=5$: **31** dan

5. Contoh principal non-leading: $\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$

Leading selalu memuat a_{11} ; submatriks yang membuang baris/kolom pertama bukan leading.

6 · Invers matriks

$$A^{-1} = C^T / \det A$$

C = matriks kofaktor; $A A^{-1} = I$

- $\det A \neq 0 \rightarrow$ **ada invers** (tak singular). $\det A = 0 \rightarrow$ **singular**, tak ada invers

- Cari k agar singular: set $\det = 0$. $\begin{vmatrix} 2 & 3 \\ k & 6 \end{vmatrix}$
 $12 - 3k = 0 \rightarrow k = 4$

$$2 \times 2: \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ -1 & 0 \end{bmatrix} \det = -2 \rightarrow A^{-1} = \begin{bmatrix} 0 & -1 \\ -0,5 & -0,5 \end{bmatrix}$$

$$3 \times 3 \text{ slide } (\det=1): C = \begin{bmatrix} 2 & 0 & 1 \\ -3 & -1 & -1 \\ -4 & -1 & -2 \end{bmatrix} \rightarrow A^{-1} =$$

$$C^T = \begin{bmatrix} 2 & -3 & -4 \\ 0 & -1 & -1 \\ 1 & -1 & -2 \end{bmatrix}$$

- Operasi baris elementer**: $[A \mid I] \rightarrow [I \mid A^{-1}]$

7 · Sistem persamaan linier

$$Ax = b \rightarrow x = A^{-1}b$$

A tak singular $\rightarrow$ solusi unik

Slide: A di atas, $b = (-2, -6, 4)^T \rightarrow x = A^{-1}b = \mathbf{(-2, 2, -4)^T}$ (bukan Ab)

Gauss-Jordan: $[A \mid b] \rightarrow$ operasi baris $\rightarrow [I \mid x]$

- Pilih pivot kolom j, jadikan 1 ($R_p \leftarrow R_p/\text{pivot}$)
- Nolkan sisa kolom: $R_i \leftarrow R_i - a_{ij}R_p$
- Ulangi tiap kolom

Slide: $R_2 + R_1$, $R_3 - R_1 \rightarrow R_2/(-2)$, $R_1 + 2R_2$, $R_3 - R_2 \rightarrow R_3/(-0,5)$, $R_1 + 2R_3$, $R_2 + 0,5R_3 \rightarrow x = \mathbf{(-2, 2, -4)}$

8 · Polinomial: akar & faktor

- Derajat $n \rightarrow$ **n akar** (boleh berulang). r akar $\Leftrightarrow (x-r)$ faktor. $f(x) = (x-r)q(x)$, q derajat $n-1 = f \div (x-r)$
- Orde 1: $x = -a_0/a_1$ ($2x-1=0 \rightarrow 1/2$)

$$\text{orde 2: } x = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$$

- $x^2 - x - 6: (1 \pm 5)/2 = 3, -2 \rightarrow (x-3)(x+2)$
- $2x^2 - 3x - 5: (3 \pm 7)/4 = 5/2, -1 \rightarrow \Sigma x^2 = 25/4 + 1 = \mathbf{29/4}$
- $x^4 - 2x^2 + 1 = (x^2 - 1)^2$: 4 akar $(1, 1, -1, -1)$, 2 berbeda

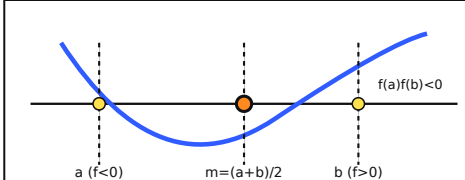
Akar rasional: p faktor a_0 , q faktor a_n

(koefisien bulat) $\rightarrow$ kandidat $x = p/q$

$x^3 - 2x^2 - 5x + 6: a_0=6, a_n=1 \rightarrow \pm 1, 2, 3, 6$. Coba $x=1$: sisa 0 $\rightarrow$ hasil bagi $x^2 - x - 6 \rightarrow$ akar **1, 3, -2**

Bagi $2x^3 - 3x^2 - 11x + 6$ oleh $(x-3)$: koef 2, 3, -2, sisa 0 $\rightarrow \mathbf{2x^2 + 3x - 2}$

9 · Bisection (cari akar)



- Pilih a_0, b_0 dengan $f(a_0), f(b_0)$ **tanda berlawanan**
- $m_k = (a_k + b_k)/2$. Jika $f(m) = 0 \rightarrow x = m$
- $f(a_k)f(m_k) > 0 \rightarrow a_{k+1} = m$, b tetap; selain itu $b_{k+1} = m$
- Henti jika $|a_k - b_k| < \epsilon$; $x = a - (b-a)f(a)/(f(b)-f(a))$

k	a	b	m	f(m)
0	1	4	2,5	-2,25
1	2,5	4	3,25	1,3125
2	2,5	3,25	2,875	-0,609

$f = x^2 - x - 6$, $[1,4]$: 12 iterasi $\rightarrow x = 3,0000$. $x^3 - 2x - 5$ pada $[2,3]$: $m = 2,5 \rightarrow [2;2,5]$, lalu $[2;2,25]$, $[2;2,125]$

10 · Newton-Raphson & perbandingan

$$x_{k+1} = x_k - f(x_k) / f'(x_k)$$

berhenti jika $|x_{k+1} - x_k| < \epsilon$

k	x	f	f'
0	4	6	7
1	3,142857	0,7347	5,2857
2	3,003861	0,0193	5,0077
3	3,000003		

$f = x^2 - x - 6$, $f' = 2x - 1$, $x_0 = 4$, $\epsilon = 0,001 \rightarrow 4$ iterasi, $x = 3$

$f = x^2 - 10$, $x_0 = 3$: $x_1 = 3 - (-1)/6 = 3,1667 \rightarrow x_2 \approx 3,1623$

	Bisection	Newton-Raphson
Butuh	a, b tanda f berlawanan	$f(x)$ & satu titik awal x_0
Turunan	tidak	ya
Contoh slide	12 iterasi	4 iterasi

Newton-Raphson (akar) memakai f/f' ; jangan tertukar dengan Newton optimisasi (f'/f , Materi 6).